MANTHAN SHRIRAM JOSHI

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Summary

Brings curiosity, structure, and adaptability to every project, with experience troubleshooting experiments and coordinating teams. Focused on problem-solving and process improvement to support collective success. Actively seeking a summer internship or a 6–8 month co-op..

Education

Northeastern University, Boston | MA | ***Master of Science in Cell and Gene Therapies*** — *December 2026*
Coursework: Stem cells and Regeneration, Cell and Gene Therapies, Immunology, Regulatory Landscapes, Advance Drug Delivery Systems, Bioinformatics — GPA: 3.7/4

Manipal Institute of Technology (MAHE), Manipal| India |***Bachelor of Technology in Biotechnology*** **| Minor: Pharmaceutical Biotechnology** — *July 2023*
Coursework: Genetics, Immunology, Bioinformatics, Pharmaceutical Science, Microbiology, Biochemistry, Bioprocess Engineering

Harvard Business School, Boston | MA | ***Certificate in Management Essentials*** — *August 2025*
8-week, 35-hour program emphasizing decision-making, implementation, organizational learning, and change management

Skills

Laboratory Techniques	CRISPR/Cas9 genome editing (knock-in/knock-out), INDEL analysis, mammalian cell culture (growth, thawing, cryopreservation), ELISA, protein purification, bacterial transformation, plasmid expansion and purification, restriction enzyme digestion, transfection, fluorescence imaging, genomic DNA isolation, sequencing validation, drug selection assays, PCR, SDS-PAGE, Western blotting
Programming Languages	Python, R Programming (beginner)

Professional Experience

Northeastern University, *Teaching Assistant - Cell and Gene Therapies*, Boston MA — *Jan 2026 - Apr 2026*

- Graded assignments and examinations using standardized rubrics to ensure fairness and academic consistency
- Supported classroom learning by clarifying core CGT concepts and incorporating relevant current developments in the field

Northeastern University, *Graduate Research Project*, Boston MA — *Sept 2025 - Dec 2025*

- Performed CRISPR/Cas9 knock-in and knock-out editing in CHO cells, including design of experimental workflow, preparation of Cas9–RNP complexes, and transfection using CRISPRmax for targeted modification of the HPRT1 locus
- Conducted full downstream validation: genomic DNA extraction, PCR amplification, INDEL analysis, Sanger sequencing, agarose gel electrophoresis, and Nanodrop-based DNA quantification to confirm genome edits
- Executed complete plasmid engineering workflow including bacterial transformation, colony screening, plasmid purification, and restriction enzyme digest verification (NotI, EcoRV)
- Managed mammalian cell culture, including seeding, passaging, cryopreservation, and optimization of puromycin selection using a CellTiter-Glo viability assay

Reagene Biosciences Pvt Ltd, *Research Intern*, Bangalore, India — *Aug 2023 – Aug 2025*

- Contributed to 3D bioprinting projects developing human tissue models as alternatives to animal testing
- Performed molecular biology, microbiology, and protein analysis techniques including DNA purification, gel electrophoresis, Nanodrop, bacterial transformations, Western blotting, ELISA, and GCS assays
- Managed mammalian cell culture and cryopreservation to maintain high viability for bioink preparation
- Supported with investor decks, grant proposals, and market research for 3D bioprinting and non-animal testing platforms

Tranalab Pvt Ltd, Center for Human Genetics, *Undergraduate Research Intern*, Bangalore India — *Jan 2023 – Aug 2023*

- Advanced a thesis project: *Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain*, awarded first rank in the 2019 batch
- Optimized recombinant protein expression and purification methods including Ni-NTA chromatography and SDS-PAGE
- Applied molecular biology techniques to achieve high-yield protein expression
- Cultivated CHO cell lines for protein production and implemented cryogenic storage techniques

Indian Institute of Technology, *Undergraduate Research Intern*, Indore, India — *Sept 2021 – Oct 2021*

- Supported project on stem cell-based tissue growth and tumor control through protocol design, MSDS preparation, and experimental planning
- Assisted with target identification and therapeutic intervention modeling

Leadership and Volunteering

Vice President, Northeastern University, *Graduate Biotech-Bioinfo Association*, Boston, MA — *December 2024 - Present*

- Leads executive board operations, managing communications, records, and coordination across committees while supporting professional development events and member engagement
- Oversees organizational strategy, planning and executing networking events, workshops, and panel discussions to strengthen community involvement and industry exposure